Laplace Equation

Q Use separation of variables method to solve the equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

subject to the boundary condition

$$u(0,y) = u(l,y) = u(x,0) = 0$$

and $u(x,a) = \sin \frac{n\pi x}{l}$.

Ans Here we have to solve Laplace equation in two dimension with given conditions

We rewrite given conditions in simple form as

$u = u(x,y)$ function So

$\boxed{\begin{array}{l} u(x,t) \\ \text{heat} \quad \text{wave} \end{array}}$

$u(x, y)$

$u(0, y) = 0$ $\quad u(l, y) = 0$

(i) $u = 0$ when $x = 0$

(ii) $u = 0$ when $x = l$

$u(x, \infty) = 0$ condition

(iii) $u = 0$ when $y = 0$

$u(x, a) = \sin \frac{n\pi x}{l}$

$u(x, y)$

(iv) $u = \sin \frac{n\pi x}{l}$ when $y = a$

By formula of Laplace Equation

in 2D

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

By method of separation of variables solution is

$$u = (c_1 \cos px + c_2 \sin px)(c_3 e^{py} + c_4 e^{-py}) \quad \text{--- (1)}$$

Write as direct or solve case II

put condition (i) $u=0$ when $x=0$ in eqn (1) we get

$$0 = (c_1 \times 1 + c_2 \times 0) \cdot (c_3 e^{py} + c_4 e^{-py})$$

$$0 = c_1 \cdot (c_3 e^{py} + c_4 e^{-py})$$

So $\boxed{c_1 = 0}$

put condition (ii) $u=0$ when $x=l$ in eqn (1)

$0 = (c_1 \cdot \cos pl + c_2 \sin pl) \cdot (c_3 e^{py} + c_4 e^{-py})$

$0 = c_2 \sin pl \cdot (c_3 e^{py} + c_4 e^{-py})$

So $\sin pl = 0$

So $\sin pl = 0$

$\sin pl = \sin n\pi$

$pl = n\pi$

$$\boxed{p = \frac{n\pi}{l}}$$

put condition (iii) $u = 0$ when $y = 0$

in eqn ① and also put c_1, p value

$0 = \left(0 + c_2 \sin \frac{n\pi x}{l}\right) \cdot (c_3 e^{0} + c_4 e^{-0})$

$0 = c_2 \sin \frac{n\pi x}{l} \cdot (c_3 + c_4)$

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So $C_3 + C_4 = 0$

$$\boxed{C_4 = -C_3}$$

put value of $\overset{0}{C_1}$, $\overset{n\pi/l}{p}$, $\overset{-C_3}{C_4}$ in equation ① of Laplace equation solution

$$u = (C_1 \cos px + C_2 \sin px) \cdot (C_3 e^{py} + C_4 e^{-py})$$

$$u = \left(0 + C_2 \sin \frac{n\pi x}{l}\right)\left(C_3 e^{py} - C_3 e^{-py}\right)$$

$$u = C_2 C_3 \sin \frac{n\pi x}{l}\left(e^{py} - e^{-py}\right)$$

$$u = C_2 C_3 \sin \frac{n\pi x}{l}\left(e^{\frac{n\pi y}{l}} - e^{-\frac{n\pi y}{l}}\right)$$

let take new constant $C_2 C_3 = b_n$

$$u = b_n \sin \frac{n\pi x}{l}\left(e^{\frac{n\pi y}{l}} - e^{-\frac{n\pi y}{l}}\right)$$

$$\frac{e^{\theta} - e^{-\theta}}{2} = \sinh\theta$$

$$u = b_n \sin\frac{n\pi x}{l}\left(e^{\frac{n\pi y}{l}} - e^{-\frac{n\pi y}{l}}\right) \quad \text{——} (2)$$

This is solution of Laplace equation

put condition (iv)

$$u = \sin\frac{n\pi x}{l} \quad \text{when } y = a \text{ in eq}^n (2)$$

$$\cancel{\sin\frac{n\pi x}{l}} = b_n \cancel{\sin\frac{n\pi x}{l}}\left(e^{\frac{n\pi a}{l}} - e^{-\frac{n\pi a}{l}}\right)$$

$$1 = b_n\, 2\left(\frac{e^{\frac{n\pi a}{l}} - e^{-\frac{n\pi a}{l}}}{2}\right)$$

$$1 = b_n\, 2 \sinh\left(\frac{n\pi a}{l}\right)$$

$$b_n = \frac{1}{2\sinh\left(\frac{n\pi a}{l}\right)}$$

put value of b_n in equation (2)

$$u = \frac{1}{\cancel{2}\sinh\left(\frac{n\pi a}{l}\right)} \sin\left(\frac{n\pi x}{l}\right) \cdot \cancel{2}\sinh\left(\frac{n\pi y}{l}\right)$$

$$\boxed{u = \frac{\sinh\left(\frac{n\pi y}{l}\right) \cdot \sin\left(\frac{n\pi x}{l}\right)}{\sinh\left(\frac{n\pi a}{l}\right)}}$$

This is final solution of Laplace equation for given problem.